SPIN-SPIN - A PHOTOSYNTHESIZER TURNTABLE

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ABSTRACT

Record playing turntables have been present since the dawn of music recording and have been used to manipulate and create new music over the last several decades. They are a familiar interface which many can relate to: the results of scratching (physically moving a playing record back and forward) are almost as intuitive as plucking guitar strings or striking keys on a piano. This paper describes the use of the Arduino [1] microcontroller along with a selection of sensors, motors and grey-scale patterns to construct a turntable device for creating and manipulating sound that is both intuitive and visually appealing.

1. INTRODUCTION

As undergraduate electronics engineering students at Victoria University of Wellington, authors Sean Pierce and Josh Boyes have developed Spin-Spin, a new interface based on a turntable and the principles of encoders. Phototransistors provide a way to read trigidhe shade on an object above it and, via analog to digital converters and software, allow it to be turned into control signals to manipulate sound. Through this technology, a new approach to a programmable sequencer was developed in the form of a turntable, a device intuitive to many users. Spin-Spin is a motorized dual-turntable with a separate control box for altering the speeds of each turntable. The patterns of each turntable can be changed to provide variation in performance and even black and white photos can be used for noisier or more random input signals.

This paper begins with a review of related works, followed by technical review of Spin-Spin, focusing on the component choices made in its design. Section 4 presents experiments, evaluating Spin-Spin's component's performance. Section 5 examines the design and construction of the interface. The paper concludes with a discussion of performance applications of Spin-Spin.

2. RELATED RESEARCH

A number of other workers have explored alternative turntable-based interfaces. This section presents a sample of such interfaces and outlines how Spin-Spin differs from them.

Berlin-based studio The-Product created a similar interface in 2012 called 'soundmachines' [2] featuring optics and turntable technology. This interface however does not allow direct manipulation of the rotating disks and the resulting music produced is exactly timed and very computer-like. By designing Spin-Spin so the disks are directly accessible by the user, music with more abstract and asynchronous timing can be achieved.

SpinCycle [3], a colour tracking turntable installation, was presented at NIME 2006, featuring user programmable patterns via placement of fluorescent tinted plexiglass disks on top of a platter. Video analysis of the rotating platter provided data for audio-visual events. This approach offers freedom in sequencing, though at an increase in cost of materials and the greater complexity of video analysis.

In light of these related works, a human-controlled interface capable of non-metric output was deemed worthy of investigation.

3. TECHNICAL DETAILS

This section seeks to detail the technical means by which the objective of a new augmented turntable interface was fulfilled.

The interface's electronic components were sourced from Sparkfun electronics [4]. Components used in Spin-Spin included an Arduino Uno, Phototransistors, DC-Motors, Force-Sensitive-Resistors, and a Potentiometer all detailed in the rest of this section.

The Arduino microcontroller, along with its programming software, is used as the analogue to digital converter between the sensor circuitry and the computer running ChucK [6]. The analogue inputs of the Arduino are utilized to connect the sensors, via voltage dividers, giving full control from the sensors. Two pulse width modulation-enabled digital pins are used to control the motor speed.

Phototransistor sensors have two parts: an infrared-emitting LED and a phototransistor; the LED's light is received by the phototransistor and its intensity measured. This component can be used to measure both distance, or how reflective different surfaces at a constant distance are. In Spin-Spin, the sensors are used to measure a constant-distance object of varying color. Black ink absorbs much of the infrared whereas white paper reflects much of it back, providing a way to measure the shade of the grey-scale patterns above the sensors.

Spin-Spin also uses Force Sensitive Resistors (FSRs). FSRs exhibit a lower resistance when an increasing external pressure is applied. Internally, a matrix of conductive leads are brought into contact when pressure is applied, thus changing the device's resistance. Practically, these can be used in as a button whose analog voltages change with increased pressure. In Spin-Spin, a simple voltage divider circuit was used with the FSR.

In Spin-Spin, a rotary potentiometer is used to control motor speed for one of the turntables by varying the output PWM to the motor.

DC motors are the mechanical component to which the turntable platters are mounted. The motors are switched with a MOSFET. A diode was used across the motor to protect the rest of the circuit from any back voltages induced by the motor when changing speeds.

4. EXPERIMENTS

A series of evaluations of the sensors were conducted prior to the fabrication of the Spin-Spin chassis, allowing the authors to verify the suitability of the sensors.

Initial experiments of passing various printed shades of grey across the phototransistor yielded repeatable patterns of varying analog to digital data (ADC) through the Arduino analog pins. The digital range of the analogue inputs on the Arduino ranged from 0 to 1023. The printed shades a few millimeters above the sensor gave a range from approximately 40 to 1000, utilizing almost the whole available range. This confirmed the ability of the sensor to be used to measure the shade of printing.

5. CONTROLLER DESIGN

In order to produce repeatable musical patterns, it was decided a turntable system could be employed to rotate patterns over the phototransistors. Patterns with various widths and shades of grey were printed on paper then glued to old second hand store 7-inch records. Figure 1 shows these patterns.

Additionally, greyscale photographs were overlaid onto more records, allowing the images' tonal variations to be converted to audio signals.



Figure 1. Examples of patterns used on disks.

A control box was designed and built alongside the main turntable interface. Altium PCB design software was used to design and implement the circuits for the sensors on an Arduino "shield", a printed circuit board that can be plugged in on top of the Arduino board.

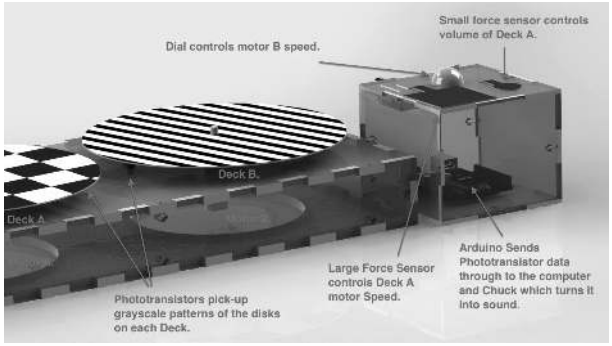


Figure 2. Turntable and Control box

DB9 cable hardware was used to connect the turntable and the controller as the cable and connectors were on hand and provided enough lines to connect the two

interfaces. The cable provided power to the turntable DC motors and access to the ADC pins on the Arduino for the phototransistors as well as necessary grounding for these components.

Figure 3 displays how the circuit was implemented on a custom milled Arduino shield. The sensors were connected to this board and along with the Arduino, were mounted inside the control box.

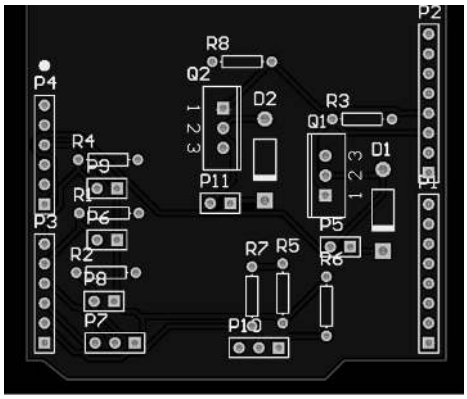


Figure 3. Control box Arduino shield.

5.1 SolidWorks Design

The enclosures for both the turntable interface and the control box were designed in the SolidWorks CAD program to be laser cut out of 6mm and 3mm acrylic and fastened together with M3 and M2 screws. This offered a high level of accuracy, which would make the boxes robust and also reduce possible axial alignment errors in the platters that held the disks (which would affect the phototransistor read values).

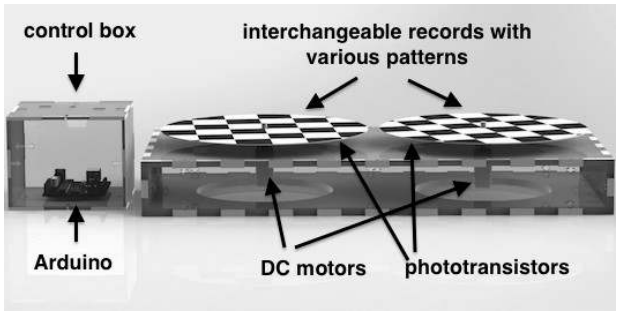


Figure 4. Rendering of the complete interface.

5.2 Sound Design

After completing structural and electronic design, Spin Spin's sound design could be explored.

All the data used for sound generation came from the phototransistor of each deck. Data from deck A controlled the frequency of a synth sound. Deck B triggered an envelope of white noise, which created a rhythm. The user could interact with these sensors in several ways. Firstly the user could actually physically spin the disks with their hands to input data via these sensors. The second way was via the other sensors on the control box. The large FSR was used to control the speed of deck A and the pot was used to control the speed of deck B. The small FSR was used to control the volume of the synth sound. Figure 5 shows the system flow of the whole interface.

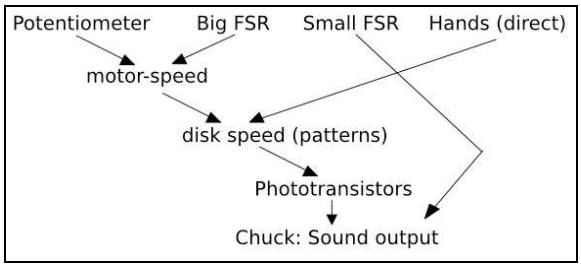


Figure 5. System flow.

5.3 Software

This section details how the ChucK and Arduino software was implemented in Spin-Spin.

The Arduino code allowed full control of the motors via the controls on the interface. The full range of the pot and large FSR were mapped to a reduced range of motor speed so the maximum and minimum settings of the sensors gave sensible extremes of motor speed. By doing this, the sensors had more control over the desired range where the audio output sounded best. Pulse width modulation from the Arduino via a MOSFET transistor circuits on the shield controlled the motor speeds as explained above in the technical section.

The Arduino code also sent data from the sensors to then be manipulated in ChucK via serial over USB.

The ChucK software design took data from the two phototransistors and the small FSR. The phototransistor of deck B triggered an envelope of white noise and a bass note square wave. This formed the rhythmic component of the musical system.

The data of the deck A phototransistor was split into ten bands over the full range. Each band corresponded to a MIDI note in an array. Several MIDI arrays were created to give different scales. The root of the scale determined

the pitch of the bass note triggered by deck B. The MIDI to frequency function was used to allow the MIDI scales to determine the pitch of a triangle wave oscillator whose gain was controlled by the small FSR.

6. MUSICAL PERFORMANCE

The Spin-Spin interface was exhibited at the 2013 New Zealand School of Music Sonic Arts Engineering Expo (SAEE). During its exhibition, viewers interacted with the sensors in a variety of ways. The familiarity of turntable-style interfaces meant the viewers could instantly connect with the device and start using it. Such intuitive control held true for the knob-style potentiometers. Audiences appeared to find the FSRs more unfamiliar: it was not immediately clear to the new user what effect they had on the outputs. Because of the common use of touch screen devices, people often thought the large square FSR was an X-Y track pad.

The ability to trigger MIDI notes rather than just using the input of the phototransistor directly as a frequency appeared to give the interface improved playability. Rather than creating an atonal sound, a tonal scale could be chosen. This functionality allows for non-musicians the ability to play in a key and mode of a scale. People of any musical skill level could interact and add to the music.



Figure 6. Spin-Spin at exhibition.

7. CONCLUSIONS AND CURRENT DEVELOPMENTS

The interface worked well, and the sonic results surpassed expectation. There is also scope to develop the interface further. Slower geared motors would enable longer loops to be made and more sturdy motor shafts would increase the accuracy of the sound produced. The audio software could also be further developed. Further exploration could be taken in using the controller to interface with Ableton Live or Max/MSP.

While the current software system is suitably functional, editing and changing parameters is not user friendly as any changes to the generated musical scales need to be made in the code. In development is a user interface built in Max/MSP [7] to allow users to: choose the number of divisions the phototransistors map to, adjust the ranges of those divisions and then translate the sensors' data from those divisions to midi note/control messages. This will allow the device to be a much more versatile tool for creating music with a computer.

In spite of the potential for future iterations, Spin-Spin represents a complete turntable-based music performance system. This system not only features an input device, but also motorized feedback and audio production facilities.

9. ACKNOWLEDGMENTS

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10. REFERENCES

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